

Sitekraft - AI-Powered Website Builder

Source Code for Copyright Registration

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Programming Language:	Python 3.12, JavaScript (React 19)
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Description of Work:

Sitekraft is a web application designed and developed by Sneha Chaudhary and Koustubh Kukreti. It takes a business description as input and generates a complete, responsive React website. The backend is written in Python using FastAPI and integrates the Google Gemini API for intelligent code generation. Stock images are fetched through the Pexels API, and the finished website is automatically deployed to Vercel. The frontend is built with React 19 and Vite, featuring dark mode support and a guided onboarding flow for users.

Note: As per Indian Copyright Office requirements for computer programs exceeding 20 pages, this document includes the first 25 pages and last 25 pages of the source code. The complete source code is available for inspection upon request.

List of Source Files

The following files form the complete original work submitted for copyright registration:

1. autonomous_agent.py - Core website generation and orchestration logic
2. autonomous_agent_react_helper.py - React code generation helper utilities
3. react_builder.py - React component builder and code assembler
4. react_template_generator.py - Dynamic page template generation engine
5. react_api_routes.py - FastAPI REST API endpoints
6. vercel_deployer.py - Vercel deployment pipeline
7. professional_copywriter.py - Content and copy generation module
8. groq_helper.py - Groq API integration
9. openrouter_helper.py - OpenRouter multi-model API integration
10. pexels_helper.py - Pexels stock image API integration
11. image_proxy.py - Image proxy and caching layer
12. supabase_config.py - Supabase database configuration
13. App.jsx - Main React frontend application
14. Onboarding.jsx - User onboarding and business input flow
15. useTheme.js - Theme management hook

Project Structure:

backend/ - Python FastAPI backend with AI and helper modules

frontend/ - React 19 + Vite frontend application

File 1: backend/autonomous_agent.py

Backend - Core AI Agent

```
1
2     from dotenv import load_dotenv
3     load_dotenv()
4     """
5     AI WEBSITE BUILDER AGENT - Interactive Chat + Full Websites
6     Conversational chatbot that guides users to build production-ready websites
7     with e-commerce, booking, media uploads, and admin panels.
8     """
9
10    import sys
11    import os
12    import json
13    import uuid
14    import random
15    import requests
16
17    # Ensure stdout uses UTF-8 on Windows (avoids cp1252 UnicodeEncodeError for arrow chars in
18    # print)
19    if hasattr(sys.stdout, 'reconfigure'):
20        sys.stdout.reconfigure(encoding='utf-8', errors='replace')
21    import zipfile
22    import io
23    import shutil
24    import base64
25    from datetime import datetime
26    from typing import Dict, Optional, List
27    import google.generativeai as genai
28    from fastapi import FastAPI, HTTPException, UploadFile, File, Form
29    from fastapi.middleware.cors import CORSMiddleware
30    from fastapi.staticfiles import StaticFiles
31    from fastapi.responses import HTMLResponse, FileResponse, StreamingResponse, JSONResponse
32    from pydantic import BaseModel
33    from autonomous_agent_react_helper import build_react_website_from_chat
34    from professional_copywriter import copywriter
35    from pexels_helper import search_pexels_image, search_pexels_images
36
37    # Import React and Vercel modules
38    try:
39        from react_builder import ReactWebsiteBuilder
40        from vercel_deployer import VercelDeployer
41        REACT_VERCEL_AVAILABLE = True
42    except ImportError as e:
43        print(f"[WARNING] React/Vercel modules not available: {e}")
44        REACT_VERCEL_AVAILABLE = False
45
46    # Load environment variables
47    GEMINI_API_KEY = os.getenv("GEMINI_API_KEY")
48
49    if not GEMINI_API_KEY:
50        raise ValueError("GEMINI_API_KEY environment variable not set")
```

```
51     genai.configure(api_key=GEMINI_API_KEY)
52
53     # Initialize FastAPI
54     app = FastAPI(title="Sitekraft Agent")
55     app.add_middleware(
56         CORSMiddleware,
57         allow_origins=["*"],
58         allow_credentials=True,
59         allow_methods=["*"],
60         allow_headers=["*"],
61     )
62
63     # Create directories
64     os.makedirs("generated_websites", exist_ok=True)
65     os.makedirs("website_states", exist_ok=True)
66     os.makedirs("static/uploads", exist_ok=True)
67     os.makedirs("site_data", exist_ok=True)
68
69     # Mount static files for uploads (used by chat image uploads)
70     app.mount("/static", StaticFiles(directory="static"), name="static")
71
72     # =====
73     # IN-MEMORY STATE
74     # =====
75     CHAT_SESSIONS = {} # session_id -> conversation state
76     WEBSITE_STATES = {} # session_id -> website state
77
78
79     # =====
80     # REQUEST MODELS
81     # =====
82     class ChatMessage(BaseModel):
83         session_id: str
84         message: str
85         uploaded_files: Optional[List[str]] = []
86
87     class BuildRequest(BaseModel):
88         description: str
89         preferences: Optional[Dict] = None
90
91     class EditRequest(BaseModel):
92         session_id: str
93         edit_request: str
94
95     class BookingRequest(BaseModel):
96         name: str
97         email: str
98         phone: str
99         service: str
100         date: str
101         time: str
102         notes: Optional[str] = ""
103
104     class OrderRequest(BaseModel):
```

```
105     name: str
106     email: str
107     phone: str
108     address: str
109     items: List[Dict]
110     notes: Optional[str] = ""
111
112     class ContactRequest(BaseModel):
113         name: str
114         email: str
115         phone: Optional[str] = ""
116         message: str
117
118
119     # =====
120     # CONVERSATION STATE MACHINE
121     # =====
122     CONVERSATION_STAGES = [
123         "greeting",          # Ask business type
124         "business_details",  # Ask name, location, description
125         "services",          # Ask about services/products
126         "media",             # Ask for photos/videos
127         "features",          # Ask what features they need
128         "style",             # Ask about colors/vibe
129         "building",          # AI is building the website
130         "review",            # Website built, user can edit via chat
131     ]
132
133     BUSINESS_TYPE_QUESTIONS = {
134         "restaurant": {
135             "services": "What type of cuisine do you serve? List your signature dishes or menu categories.",
136             "media": "Do you have photos of your restaurant or dishes? Share them and I'll add them to your website! If not, I'll find beautiful stock photos.",
137             "features": "Which features do you need?",
138             "features_options": ["Online Menu", "Table Booking", "Online Ordering", "Delivery Tracking", "Reviews Section", "Photo Gallery"],
139         },
140         "salon": {
141             "services": "What services do you offer? (e.g., Haircuts, Coloring, Manicure, Facial, etc.)",
142             "media": "Do you have photos of your salon or your work? Share them! If not, I'll use professional stock images.",
143             "features": "Which features do you need?",
144             "features_options": ["Online Booking", "Service Menu with Prices", "Before/After Gallery", "Reviews", "Gift Cards"],
145         },
146         "gym": {
147             "services": "What fitness programs do you offer? (e.g., HIIT, Yoga, CrossFit, Personal Training)",
148             "media": "Do you have photos of your gym or training sessions? Share them!",
149             "features": "Which features do you need?",
150             "features_options": ["Membership Plans", "Class Schedule", "Online Booking", "Trainer Profiles", "Transformation Gallery"],
```

```
151     },
152     "shop": {
153         "services": "What products do you sell? List your main categories.",
154         "media": "Do you have product photos? Share them and I'll showcase them
beautifully!",
155         "features": "Which features do you need?",
156         "features_options": ["Product Catalog", "Online Ordering", "Shopping Cart",
"Reviews", "Size Guide"],
157     },
158     "clinic": {
159         "services": "What medical services do you provide? (e.g., General Practice, Dental,
Dermatology)",
160         "media": "Do you have photos of your clinic or team? Share them!",
161         "features": "Which features do you need?",
162         "features_options": ["Appointment Booking", "Doctor Profiles", "Service List",
"Patient Reviews", "Insurance Info"],
163     },
164     "cafe": {
165         "services": "What do you serve? (e.g., Specialty Coffee, Pastries, Breakfast,
Lunch)",
166         "media": "Do you have photos of your cafe or menu items? Share them!",
167         "features": "Which features do you need?",
168         "features_options": ["Menu Display", "Online Ordering", "Table Reservation", "Photo
Gallery", "Loyalty Program"],
169     },
170     "default": {
171         "services": "What are your main services or products?",
172         "media": "Do you have any photos or videos of your business? Share them and I'll add
them to your website!",
173         "features": "Which features do you need?",
174         "features_options": ["Contact Form", "Service Catalog", "Online Booking", "Photo
Gallery", "Testimonials", "Blog"],
175     }
176 }
177
178
179 def get_business_questions(business_type):
180     btype = business_type.lower()
181     for key in BUSINESS_TYPE_QUESTIONS:
182         if key in btype:
183             return BUSINESS_TYPE_QUESTIONS[key]
184     return BUSINESS_TYPE_QUESTIONS["default"]
185
186
187 # =====
188 # IMAGE SEARCH ENDPOINT
189 # =====
190
191 @app.get("/api/image-search")
192 async def image_search(query: str, orientation: str = ""):
193     """Search Pexels and return the direct image URL.
194     Direct Pexels URLs work fine in <img> tags and CSS url() on any page (blob, Vercel,
etc.).
195     Returning the URL directly keeps the HTML tiny, makes deployment reliable, and is ~10x
```

```
faster.
196     """
197     pexels_url = search_pexels_image(query, orientation)
198     print(f"[image-search] '{query}' -> {(pexels_url or 'none')[:80]}")
199     return {"url": pexels_url}
200
201
202 # =====
203 # CHAT ENDPOINTS
204 # =====
205
206 @app.post("/api/chat/start")
207 async def start_chat():
208     """Start a new chat session"""
209     session_id = str(uuid.uuid4())
210     CHAT_SESSIONS[session_id] = {
211         "stage": "greeting",
212         "business_type": None,
213         "business_name": None,
214         "location": None,
215         "description": None,
216         "services": [],
217         "uploaded_media": [],
218         "features": [],
219         "style_vibe": None,
220         "conversation_history": [],
221         "website_session_id": None,
222     }
223
224     return {
225         "session_id": session_id,
226         "message": "Hey there! I'm your AI website builder. I'll help you create a stunning,
227         fully functional website for your business.\n\nWhat kind of business do you have?",
228         "quick_replies": ["Restaurant", "Salon / Spa", "Gym / Fitness", "Online Shop",
229         "Clinic / Healthcare", "Cafe / Coffee", "Other"]
230     }
231
232 @app.post("/api/chat")
233 async def chat(request: ChatMessage):
234     """Process a chat message and return AI response"""
235     session_id = request.session_id
236     message = request.message.strip()
237     uploaded_files = request.uploaded_files or []
238
239     if session_id not in CHAT_SESSIONS:
240         # Auto-create session if missing
241         CHAT_SESSIONS[session_id] = {
242             "stage": "greeting",
243             "business_type": None,
244             "business_name": None,
245             "location": None,
246             "description": None,
247             "services": [],
```

```
247         "uploaded_media": [],
248         "features": [],
249         "style_vibe": None,
250         "conversation_history": [],
251         "website_session_id": None,
252     }
253
254     session = CHAT_SESSIONS[session_id]
255     stage = session["stage"]
256
257     # Store uploaded files
258     if uploaded_files:
259         session["uploaded_media"].extend(uploaded_files)
260
261     # Add to conversation history
262     session["conversation_history"].append({"role": "user", "content": message, "files":
uploaded_files})
263
264     response_data = {"session_id": session_id, "preview_ready": False, "website_code": "",
"quick_replies": []}
265
266     try:
267         # ---- GREETING STAGE: Identify business type ----
268         if stage == "greeting":
269             business_type = identify_business_type(message)
270             session["business_type"] = business_type
271             session["stage"] = "business_details"
272
273             response_data["message"] = (
274                 f"Great, a **{business_type}** business! Let me help you build an amazing
website.\n\n"
275                 f"Tell me a bit more:\n"
276                 f"- What's your business name?\n"
277                 f"- Where is it located (city)?\n"
278                 f"- A brief description of what makes it special"
279             )
280
281         # ---- BUSINESS DETAILS STAGE ----
282         elif stage == "business_details":
283             # Use AI to extract business details
284             details = extract_business_details(message, session["business_type"])
285             session["business_name"] = details.get("name", "My Business")
286             session["location"] = details.get("location", "")
287             session["description"] = message
288             session["stage"] = "services"
289
290             questions = get_business_questions(session["business_type"])
291             response_data["message"] = (
292                 f"**{session['business_name']}** sounds wonderful!\n\n"
293                 f"{questions['services']}"
294             )
295
296         # ---- SERVICES STAGE ----
297         elif stage == "services":
```



```
298         services = extract_services(message, session["business_type"])
299         session["services"] = services
300         session["stage"] = "media"
301
302         questions = get_business_questions(session["business_type"])
303         response_data["message"] = (
304             f"Noted! I'll feature these on your website.\n\n"
305             f"{questions['media']}\n\n"
306             f"You can upload photos using the **** button, or type **skip** to use stock
images."
307         )
308         response_data["quick_replies"] = ["Skip - use stock photos", "I'll upload some
photos"]
309
310         # ---- MEDIA STAGE ----
311         elif stage == "media":
312             if uploaded_files:
313                 count = len(session["uploaded_media"])
314                 response_data["message"] = (
315                     f"Got {count} photo(s)! They'll look great on your website.\n\n"
316                     f"Want to add more, or shall we move on to features?"
317                 )
318                 response_data["quick_replies"] = ["Add more photos", "Move on to features"]
319                 # Stay in media stage if they want to add more
320                 if "more" not in message.lower() and "add" not in message.lower():
321                     session["stage"] = "features"
322                     questions = get_business_questions(session["business_type"])
323                     response_data["message"] = (
324                         f"Got {count} photo(s)! They'll look great on your website.\n\n"
325                         f"{questions['features']}"
326                     )
327                     response_data["quick_replies"] = questions["features_options"]
328                     response_data["multi_select"] = True
329             else:
330                 if "skip" in message.lower() or "stock" in message.lower() or "no" in
message.lower() or "don't" in message.lower():
331                     session["stage"] = "features"
332                     questions = get_business_questions(session["business_type"])
333                     response_data["message"] = (
334                         f"No worries! I'll find beautiful professional photos for you.\n\n"
335                         f"{questions['features']}"
336                     )
337                     response_data["quick_replies"] = questions["features_options"]
338                     response_data["multi_select"] = True
339                 elif "upload" in message.lower() or "photo" in message.lower() or "i'll" in
message.lower():
340                     response_data["message"] = "Go ahead! Click the **** button to upload
your photos. You can upload multiple at once."
341                 else:
342                     session["stage"] = "features"
343                     questions = get_business_questions(session["business_type"])
344                     response_data["message"] = (
345                         f"Got it! Moving on.\n\n{questions['features']}"
346                     )
```

```
347         response_data["quick_replies"] = questions["features_options"]
348         response_data["multi_select"] = True
349
350     # ---- FEATURES STAGE ----
351     elif stage == "features":
352         features = extract_features(message, session["business_type"])
353         session["features"] = features
354         session["stage"] = "style"
355
356         response_data["message"] = (
357             f"Great choices! Last question - what style/vibe do you want for your
website?"
358         )
359         response_data["quick_replies"] = ["Modern & Clean", "Luxury & Elegant", "Warm &
Cozy", "Bold & Energetic", "Minimalist", "Let AI decide"]
360
361     # ---- STYLE STAGE ----
362     elif stage == "style":
363         session["style_vibe"] = message.lower().replace("let ai decide", "modern")
364         session["stage"] = "building"
365
366         response_data["message"] = (
367             f"Perfect! I have everything I need. Let me build your website now...\n\n"
368             f"Building your {session['business_type']} website for
{session['business_name']}...\n\n"
369             f"Hold on while I create something amazing..."
370         )
371
372     # Actually build the React website
373     try:
374         import uuid
375         session["website_session_id"] = str(uuid.uuid4())[:8]
376         build_result = await build_react_website_from_chat(session)
377         website_session_id = build_result["session_id"]
378         session["website_session_id"] = website_session_id
379         session["stage"] = "review"
380
381         response_data["session_id"] = website_session_id
382         response_data["preview_ready"] = True
383         response_data["website_code"] = build_result["html"]
384
385         # Get Vercel URL from build result
386         vercel_url = build_result.get("vercel_url")
387         if vercel_url:
388             session["vercel_url"] = vercel_url
389             print(f"[OK] Deployed to Vercel: {vercel_url}")
390
391         message_text = f"\n\nYour website is ready! Check the preview on the
right.\n\n"
392         if vercel_url:
393             message_text += f"Live Link: {vercel_url}\n(Your website is live and
shareable!)\n\n"
394         message_text += "Want to make any changes? Just tell me what you'd like to
modify - colors, text, layout, anything!"
```

```
395
396         response_data["message"] += message_text
397         response_data["vercel_url"] = vercel_url
398         response_data["quick_replies"] = ["Change colors", "Edit text", "Add more
sections", "Looks perfect!", "Download it"]
399
400     except Exception as e:
401         print(f"Build error: {e}")
402         session["stage"] = "style"
403         response_data["message"] = f"Sorry, I hit an issue building your website. Let
me try again. Error: {str(e)[:100]}"
404
405     # ---- REVIEW/EDIT STAGE ----
406     elif stage == "review":
407         if "download" in message.lower():
408             response_data["message"] = "You can download your website using the
**Download** button in the preview panel!"
409             response_data["quick_replies"] = ["Change colors", "Edit text", "Add more
sections", "Start over"]
410             elif "perfect" in message.lower() or "great" in message.lower() or "love it" in
message.lower() or "looks good" in message.lower():
411                 response_data["message"] = (
412                     "Awesome! Your website is ready to go.\n\n"
413                     "You can:\n"
414                     "- **Download** it using the button in the preview panel\n"
415                     "- **Open in new tab** to see the full version\n"
416                     "- Keep chatting with me to make more changes\n\n"
417                     "Need anything else?"
418                 )
419                 response_data["quick_replies"] = ["Make more changes", "Download", "Start a
new website"]
420             elif "start over" in message.lower() or "new website" in message.lower():
421                 # Reset session
422                 new_session = await start_chat()
423                 return new_session
424             else:
425                 # Apply edit via AI
426                 try:
427                     edit_result = edit_website_from_chat(session, message)
428                     session["website_session_id"] = edit_result["session_id"]
429
430                     response_data["session_id"] = edit_result["session_id"]
431                     response_data["preview_ready"] = True
432                     response_data["website_code"] = edit_result["html"]
433                     response_data["message"] = (
434                         f"Done! I've applied your changes. Check the updated preview.\n\n"
435                         f"Want to change anything else?"
436                     )
437                     response_data["quick_replies"] = ["Change colors", "Edit text", "Add
sections", "Looks perfect!", "Download"]
438
439                 except Exception as e:
440                     print(f"Edit error: {e}")
441                     response_data["message"] = f"Sorry, I couldn't apply that change. Could
```

```
        you try rephrasing? Error: {str(e)[:100]}"
442
443         # ---- BUILDING STAGE (shouldn't normally reach here) ----
444         elif stage == "building":
445             response_data["message"] = "I'm still building your website. Please wait a
moment..."
446
447         except Exception as e:
448             print(f"Chat error: {e}")
449             response_data["message"] = "Sorry, something went wrong. Could you try again?"
450
451         # Store AI response in history
452         session["conversation_history"].append({"role": "assistant", "content":
response_data["message"]})
453
454         return response_data
455
456
457     # =====
458     # AI HELPER FUNCTIONS
459     # =====
460
461     def identify_business_type(message):
462         """Identify business type from user message"""
463         msg = message.lower()
464         type_map = {
465             "restaurant": ["restaurant", "food", "dining", "eatery", "bistro", "diner"],
466             "salon": ["salon", "spa", "beauty", "hair", "nail", "barber"],
467             "gym": ["gym", "fitness", "workout", "training", "crossfit", "yoga"],
468             "shop": ["shop", "store", "ecommerce", "e-commerce", "retail", "boutique",
"clothing", "fashion"],
469             "clinic": ["clinic", "hospital", "doctor", "medical", "dental", "healthcare",
"health"],
470             "cafe": ["cafe", "coffee", "bakery", "tea", "pastry"],
471         }
472         for btype, keywords in type_map.items():
473             if any(kw in msg for kw in keywords):
474                 return btype
475
476         # Use AI for ambiguous cases
477         try:
478             model = genai.GenerativeModel('gemini-2.5-flash')
479             resp = model.generate_content(
480                 f'What type of business is this? "{message}". '
481                 f'Reply with ONLY one word from: restaurant, salon, gym, shop, clinic, cafe,
business'
482             )
483             return resp.text.strip().lower().replace('.', '')
484         except:
485             return "business"
486
487
488     def extract_business_details(message, business_type):
489         """Extract business name and location from message"""
```

```
490     try:
491         model = genai.GenerativeModel('gemini-2.5-flash')
492         resp = model.generate_content(
493             f'Extract the business name and city/location from this text: "{message}". '
494             f'The business type is: {business_type}. '
495             f'Return ONLY valid JSON with no extra text: {{"name": "exact business name from
text", "location": "city, country"}}'
496         )
497         text = resp.text.strip()
498         json_start = text.find('{')
499         json_end = text.rfind('}') + 1
500         if json_start >= 0:
501             result = json.loads(text[json_start:json_end])
502             print(f"Extracted details: {result}")
503             return result
504     except Exception as e:
505         print(f"Extract details error: {e}")
506
507     # Fallback: try to find name in the message itself
508     msg_lower = message.lower()
509     name = "My Business"
510     # Simple heuristic: look for "name is X" or "called X"
511     for pattern in ["name is ", "name's ", "called ", "named "]:
512         if pattern in msg_lower:
513             idx = msg_lower.index(pattern) + len(pattern)
514             rest = message[idx:].strip()
515             # Take until next punctuation or common word
516             for end_word in [" i ", " we ", " in ", " at ", " located", " based", ",", ".", ":"]:
517                 if end_word in rest.lower():
518                     name = rest[:rest.lower().index(end_word)].strip()
519                     break
520             else:
521                 name = rest.split()[0] if rest else "My Business"
522             break
523     return {"name": name, "location": ""}
524
525
526 def extract_services(message, business_type):
527     """Extract services/products from message"""
528     try:
529         model = genai.GenerativeModel('gemini-2.5-flash')
530         resp = model.generate_content(
531             f'Extract the specific product categories or services mentioned in this text:
"{message}". '
532             f'Business type: {business_type}. '
533             f'Return ONLY a JSON array of the items mentioned, keep the exact names: ["Item
1", "Item 2", ...]'
534         )
535         text = resp.text.strip()
536         json_start = text.find('[')
537         json_end = text.rfind(']') + 1
538         if json_start >= 0:
539             result = json.loads(text[json_start:json_end])
540             if result and len(result) > 0:
```

```
541         print(f"Extracted services: {result}")
542         return result
543     except Exception as e:
544         print(f"Extract services error: {e}")
545
546     # Fallback: split by commas and common separators
547     parts = []
548     for sep in [",", " - ", "- ", " and ", "\n", "/"]:
549         if sep in message:
550             parts = [p.strip() for p in message.split(sep) if p.strip() and len(p.strip()) >
1]
551             break
552     if parts:
553         return parts[:8]
554     return [message.strip()] if message.strip() else ["Products"]
555
556
557 def extract_features(message, business_type):
558     """Extract requested features from message - captures ALL selected features"""
559     features = []
560     msg = message.lower()
561
562     # Comprehensive feature mapping - maps user selections to internal feature names
563     feature_mappings = {
564         # Booking related
565         "booking": ["booking", "appointment", "schedule", "reserve", "online booking", "table
booking"],
566
567         # Ordering/Shopping related
568         "ordering": ["order", "cart", "buy", "purchase", "delivery", "online ordering",
"shopping cart"],
569
570         # Menu/Catalog related
571         "menu": ["menu", "catalog", "product", "items", "service menu", "product catalog",
"class schedule"],
572
573         # Gallery related
574         "gallery": ["gallery", "photos", "portfolio", "before/after", "photo gallery",
"transformation gallery"],
575
576         # Reviews related
577         "reviews": ["review", "testimonial", "feedback", "patient reviews"],
578
579         # Contact related
580         "contact": ["contact", "form", "inquiry", "enquiry", "contact form"],
581
582         # Blog/Content related
583         "blog": ["blog", "news", "articles"],
584
585         # Membership/Plans
586         "membership": ["membership", "plans", "pricing", "packages", "membership plans"],
587
588         # Profiles
589         "profiles": ["profiles", "team", "trainers", "doctors", "trainer profiles", "doctor
```

```
        profiles"],
590
591        # Tracking
592        "tracking": ["tracking", "delivery tracking", "order tracking"],
593
594        # Size guide
595        "size_guide": ["size", "sizing", "size guide", "size chart"],
596
597        # Gift cards
598        "gift_cards": ["gift", "cards", "gift cards", "vouchers"],
599
600        # Loyalty
601        "loyalty": ["loyalty", "rewards", "loyalty program"],
602
603        # Insurance
604        "insurance": ["insurance", "insurance info"],
605    }
606
607    # Check each feature mapping
608    for feature, keywords in feature_mappings.items():
609        if any(kw in msg for kw in keywords):
610            if feature not in features: # Avoid duplicates
611                features.append(feature)
612
613    # If still no features found, use defaults
614    if not features:
615        defaults = {
616            "restaurant": ["menu", "ordering", "reviews", "gallery", "contact"],
617            "salon": ["booking", "gallery", "reviews", "contact"],
618            "gym": ["booking", "membership", "gallery", "reviews", "contact"],
619            "shop": ["menu", "ordering", "reviews", "gallery", "contact"],
620            "store": ["menu", "ordering", "reviews", "gallery", "contact"],
621            "ecommerce": ["menu", "ordering", "reviews", "gallery", "contact"],
622            "boutique": ["menu", "ordering", "reviews", "gallery", "contact"],
623            "clinic": ["booking", "profiles", "reviews", "contact"],
624            "cafe": ["menu", "ordering", "gallery", "reviews", "contact"],
625        }
626        features = defaults.get(business_type, ["contact", "gallery", "reviews"])
627
628    print(f"[Features] Extracted from '{message}': {features}")
629    return features
630
631
632    # =====
633    # PEXELS IMAGE FETCHING
634    # =====
635
636    def fetch_pexels_images(query, count=1):
637        """Fetch images from Pexels API"""
638        urls = search_pexels_images(query, count=count, orientation="landscape")
639        if urls:
640            return urls
641        # picsum fallback if Pexels fails
642        print(f"[Pexels] Using fallback images for '{query}'")
```

```
643         return [f"https://picsum.photos/seed/{query.replace(' ', '')}/1920/1080" for _ in
range(count)]
644
645
646 def get_images_for_website(session):
647     """Get images - use uploaded ones first, fill with Pexels (all fetches run in
parallel)"""
648     from concurrent.futures import ThreadPoolExecutor, as_completed
649
650     uploaded = session.get("uploaded_media", [])
651     business_type = session.get("business_type", "business")
652     business_name = session.get("business_name", "")
653     services = session.get("services", [])
654
655     images = {"hero": "", "about": "", "services": [], "gallery": [], "products": {}}
656
657     # Uploaded images become gallery/service images
658     uploaded_urls = [f"{u}" if u.startswith("http") else u for u in uploaded]
659
660     # Specific image queries for better results across different business types
661     service_queries = {
662         # E-commerce / Fashion - Western
663         "jeans": "women jeans denim fashion stylish",
664         "tops": "women tops blouse fashion trendy",
665         "top": "women top blouse fashion casual",
666         "dresses": "women dress fashion elegant",
667         "dress": "women dress fashion elegant",
668         "skorts": "women skort fashion active trendy",
669         "skort": "women skort fashion active trendy",
670         "skirts": "women skirt fashion midi maxi",
671         "skirt": "women skirt fashion midi maxi",
672         "bell bottom": "women bell bottom pants flare jeans retro",
673         "bell-bottom": "women bell bottom pants flare jeans retro",
674         "palazzo": "women palazzo pants wide leg fashion",
675         "trousers": "women trousers formal pants fashion",
676         "shorts": "women shorts casual fashion summer",
677         "co-ord": "women co-ord set matching outfit",
678         "coord": "women co-ord set matching outfit",
679         "handbags": "women handbag purse fashion luxury",
680         "accessories": "women fashion accessories jewelry",
681         "shoes": "women shoes heels fashion footwear",
682         "jackets": "women jacket fashion outerwear blazer",
683         "jacket": "women jacket fashion outerwear",
684         "sweaters": "women sweater knitwear fashion cozy",
685         "sweater": "women sweater knitwear fashion",
686         "activewear": "women activewear gym fitness sportswear",
687         "lingerie": "women lingerie innerwear",
688         "swimwear": "women swimwear beach fashion",
689
690         # E-commerce / Fashion - Indian Ethnic
691         "kurta": "women kurta kurti ethnic indian fashion",
692         "kurti": "women kurta kurti ethnic indian fashion",
693         "saree": "women saree sari indian ethnic fashion",
694         "sarree": "women saree sari indian ethnic fashion",
```



```
695     "sari": "women saree sari indian ethnic fashion",
696     "lehenga": "women lehenga bridal indian wedding fashion",
697     "lehnga": "women lehenga bridal indian wedding fashion",
698     "suits": "women salwar suit punjabi ethnic indian",
699     "suit": "women salwar suit punjabi ethnic indian",
700     "salwar": "women salwar kameez suit indian ethnic",
701     "anarkali": "women anarkali suit indian ethnic fashion",
702     "indo-western": "women indo western fusion indian fashion",
703     "indo western": "women indo western fusion indian fashion",
704     "dupatta": "women dupatta scarf indian ethnic",
705     "churidar": "women churidar pants indian ethnic",
706     "palazzo suit": "women palazzo suit indian ethnic fusion",
707
708     # Salon / Spa / Beauty
709     "facial": "woman facial spa treatment skincare glowing",
710     "massage": "spa massage therapy relaxation wellness luxury",
711     "body massage": "spa body massage therapy wellness relax",
712     "hair styling": "hairstylist salon hair beauty styling",
713     "hair cut": "hairstylist salon haircut styling modern",
714     "haircut": "hairstylist salon haircut styling",
715     "bridal makeup": "bridal makeup wedding beauty bride",
716     "makeup": "makeup artist beauty cosmetics professional",
717     "manicure": "manicure nails beauty salon polish",
718     "pedicure": "pedicure feet spa treatment nails",
719     "waxing": "beauty salon spa waxing treatment",
720     "hair color": "hair coloring salon hairstylist dye",
721     "spa": "luxury spa wellness relaxation massage",
722     "beauty": "beauty salon treatment skincare cosmetics",
723     "skincare": "skincare facial beauty treatment serum",
724     "micro facial": "facial skincare microdermabrasion beauty spa",
725     "threading": "eyebrow threading beauty salon",
726     "bleach": "facial bleach beauty salon treatment",
727     "cleanup": "facial cleanup beauty salon skincare",
728     "tan removal": "tan removal beauty treatment skincare",
729
730     # Fitness / Gym
731     "hiit": "HIIT workout fitness gym training intense",
732     "yoga": "yoga meditation fitness wellness peaceful",
733     "crossfit": "crossfit gym workout training athletes",
734     "personal training": "personal trainer gym fitness coaching",
735     "gym": "modern fitness gym workout equipment",
736     "fitness": "fitness training workout gym active",
737     "cardio": "cardio workout fitness gym running",
738     "strength": "strength training weights gym muscle",
739     "strength training": "strength training weights gym dumbbells",
740     "pilates": "pilates workout fitness studio mat",
741     "zumba": "zumba dance fitness class energetic",
742     "aerobics": "aerobics fitness class workout group",
743     "spinning": "spinning bike cycling fitness gym",
744     "weight training": "weight training gym dumbbells fitness",
745     "bodybuilding": "bodybuilding gym muscles fitness",
746     "martial arts": "martial arts karate fitness training",
747     "boxing": "boxing gym fitness training punching",
748     "dance": "dance fitness class studio energetic",
```

```
749
750     # Restaurant / Cafe / Food
751     "pizza": "pizza italian food restaurant delicious",
752     "pasta": "pasta italian cuisine food restaurant",
753     "burger": "burger food restaurant gourmet delicious",
754     "sushi": "sushi japanese food restaurant fresh",
755     "dessert": "dessert pastry cake bakery sweet",
756     "coffee": "coffee cafe espresso latte barista",
757     "salad": "fresh salad healthy food bowl colorful",
758     "steak": "steak meat grill restaurant juicy",
759     "seafood": "seafood fish restaurant cuisine fresh",
760     "wine": "wine bottle glass restaurant elegant",
761     "cocktail": "cocktail drink bar mixology colorful",
762     "breakfast": "breakfast food cafe morning pancakes",
763     "lunch": "lunch food restaurant meal delicious",
764     "dinner": "dinner food restaurant meal elegant",
765     "brunch": "brunch food cafe breakfast lunch",
766     "appetizers": "appetizers food restaurant starter",
767     "sandwiches": "sandwich food cafe fresh deli",
768     "wraps": "wraps food healthy lunch fresh",
769     "smoothies": "smoothie bowl healthy fruit fresh",
770     "juice": "fresh juice fruit healthy colorful",
771     "ice cream": "ice cream dessert sweet colorful",
772     "donuts": "donuts pastry bakery sweet glazed",
773     "cupcakes": "cupcakes bakery sweet frosting colorful",
774     "cookies": "cookies bakery sweet homemade",
775     "waffles": "waffles breakfast cafe sweet syrup",
776     "crepes": "crepes french thin pancake sweet cafe",
777     "pancakes": "pancakes breakfast fluffy syrup stack",
778     "french fries": "french fries crispy fast food snack",
779     "fries": "french fries crispy fast food snack",
780     "nachos": "nachos cheese snack mexican food",
781     "noodles": "noodles bowl asian food restaurant",
782     "fried rice": "fried rice asian food restaurant wok",
783     "rice": "rice dish food restaurant bowl",
784     "soup": "soup bowl hot food restaurant",
785     "chicken": "chicken dish grilled food restaurant",
786     "mutton": "mutton curry dish food restaurant",
787     "lamb": "lamb chops grilled food restaurant",
788     "fish": "fish fillet grilled food restaurant",
789     "prawns": "prawns shrimp seafood dish food",
790     "tacos": "tacos mexican food restaurant fresh",
791     "sandwich": "sandwich food cafe fresh deli",
792     "pasta salad": "pasta salad food fresh healthy",
793     "milkshake": "milkshake creamy thick drink sweet",
794     "mocktail": "mocktail colorful drink fresh fruit",
795     "lassi": "lassi yogurt drink indian refreshing",
796     "tea": "tea cup hot drink cafe",
797     "chai": "masala chai tea indian hot drink",
798
799     # Indian Food
800     "biryani": "biryani rice aromatic indian food",
801     "curry": "curry indian food spicy bowl",
802     "dal": "dal lentil soup indian food",
```

```
803     "dal makhani": "dal makhani indian food creamy",
804     "naan": "naan bread indian restaurant flatbread",
805     "roti": "roti chapati indian bread food",
806     "chapati": "chapati roti indian bread food",
807     "tandoori": "tandoori chicken indian food grill",
808     "tikka": "chicken tikka indian food grill spicy",
809     "tikka masala": "chicken tikka masala indian curry food",
810     "butter chicken": "butter chicken makhani indian curry",
811     "paneer": "paneer tikka indian food restaurant",
812     "dosa": "dosa south indian food crispy",
813     "idli": "idli sambar south indian food",
814     "samosa": "samosa indian snack fried crispy",
815     "chaat": "chaat indian street food snack",
816     "pav bhaji": "pav bhaji indian street food",
817     "chole": "chole bhature indian food chickpea",
818     "rajma": "rajma chawal indian food kidney beans",
819     "gulab jamun": "gulab jamun indian sweet dessert",
820     "rasgulla": "rasgulla indian sweet dessert syrup",
821     "kheer": "kheer rice pudding indian dessert",
822     "halwa": "halwa indian sweet dessert",
823     "kulfi": "kulfi indian ice cream dessert",
824     "ladoo": "ladoo indian sweet dessert ball",
825     "barfi": "barfi indian sweet dessert milk",
826     "jalebi": "jalebi indian sweet dessert crispy",
827
828     # Bakery / Pastry
829     "bread": "fresh bread bakery artisan loaves",
830     "croissant": "croissant pastry bakery french flaky",
831     "croissants": "croissants pastry bakery french",
832     "pastries": "pastries bakery sweet danish croissant",
833     "pastry": "pastry bakery sweet delicate",
834     "cakes": "cakes bakery celebration birthday wedding",
835     "cake": "cake bakery celebration frosting layers",
836     "cheesecake": "cheesecake dessert creamy slice",
837     "cheese cake": "cheesecake dessert creamy slice",
838     "brownies": "brownies chocolate bakery fudge",
839     "muffins": "muffins bakery breakfast blueberry",
840     "tarts": "tarts pastry bakery fruit elegant",
841     "pies": "pies bakery fruit dessert homemade",
842     "macarons": "macarons pastry bakery french colorful",
843     "eclairs": "eclairs pastry bakery french chocolate",
844     "baguette": "baguette bread bakery french artisan",
845     "custom cakes": "custom cake bakery celebration decorated",
846     "wedding cakes": "wedding cake bakery elegant tiered",
847     "birthday cakes": "birthday cake bakery celebration colorful",
848
849     # Salon / Spa extras
850     "nail extensions": "nail extensions acrylic nails beauty salon",
851     "gel nails": "gel nails manicure beauty salon",
852     "nail art": "nail art design beauty salon colorful",
853     "eyelash extensions": "eyelash extensions beauty salon glam",
854     "keratin treatment": "keratin hair treatment salon smooth",
855     "hair treatment": "hair treatment salon spa deep conditioning",
856     "hair spa": "hair spa treatment salon nourishing",
```

```
857     "balayage": "balayage hair color salon highlights",
858     "highlights": "hair highlights color salon styling",
859     "eyebrow shaping": "eyebrow shaping beauty salon grooming",
860     "pre-bridal": "pre-bridal package beauty salon bride",
861     "bridal package": "bridal makeup package salon bride",
862     "detan": "tan removal detan beauty treatment face",
863
864     # Fitness extras
865     "swimming": "swimming pool fitness workout laps",
866     "kickboxing": "kickboxing martial arts fitness training",
867     "meditation": "meditation yoga peaceful mindfulness wellness",
868     "stretching": "stretching flexibility yoga fitness",
869     "nutrition": "nutrition healthy food diet wellness",
870     "nutrition counseling": "nutrition counseling diet healthy food",
871     "functional training": "functional training workout fitness gym",
872     "group fitness": "group fitness class gym workout energetic",
873     "cycling": "cycling bike fitness workout gym",
874
875     # Clinic / Healthcare
876     "physiotherapy": "physiotherapy physical therapy rehabilitation exercise",
877     "dermatology": "dermatology skin treatment clinic doctor",
878     "cardiology": "cardiology heart care clinic doctor",
879     "pediatrics": "pediatrics children health clinic doctor",
880     "gynecology": "gynecology women health clinic doctor",
881     "orthopedics": "orthopedics bone joint clinic doctor",
882     "ophthalmology": "ophthalmology eye care clinic doctor",
883     "ayurveda": "ayurveda herbal medicine treatment wellness",
884     "homeopathy": "homeopathy medicine treatment clinic",
885     "dentist": "dentist dental clinic teeth care professional",
886     "doctor": "doctor clinic healthcare professional stethoscope",
887     "general practice": "doctor general practice clinic healthcare",
888     "blood test": "blood test laboratory clinic healthcare",
889     "x-ray": "x-ray radiology clinic medical imaging",
890     "ultrasound": "ultrasound scan clinic medical",
891
892     # Men's Fashion / Clothing
893     "shirts": "men shirts fashion formal casual",
894     "shirt": "men shirt fashion formal casual",
895     "t-shirts": "men t-shirt casual fashion streetwear",
896     "t-shirt": "men t-shirt casual fashion streetwear",
897     "men jeans": "men jeans denim fashion casual",
898     "men kurta": "men kurta ethnic indian fashion",
899     "sherwani": "men sherwani wedding indian ethnic fashion",
900     "blazer": "men blazer suit formal fashion",
901     "formal wear": "men formal wear suit professional",
902     "ethnic wear": "men ethnic wear indian fashion kurta",
903
904     # Jewelry / Accessories
905     "necklace": "necklace jewelry fashion gold silver",
906     "earrings": "earrings jewelry fashion gold silver",
907     "bangles": "bangles jewelry indian fashion gold",
908     "rings": "rings jewelry fashion gold diamond",
909     "bracelet": "bracelet jewelry fashion wrist",
910     "anklet": "anklet jewelry fashion gold",
```

```
911     "jewelry": "jewelry fashion gold silver elegant",
912     "watch": "watch wristwatch fashion luxury timepiece",
913     "sunglasses": "sunglasses eyewear fashion trendy",
914     "bag": "handbag purse fashion women leather",
915     "clutch": "clutch bag fashion women evening",
916     "wallet": "wallet leather fashion men women",
917
918     # Footwear
919     "heels": "women heels shoes fashion elegant",
920     "sandals": "sandals shoes fashion summer casual",
921     "boots": "boots shoes fashion leather",
922     "sneakers": "sneakers shoes fashion casual sporty",
923     "flats": "flat shoes women fashion casual",
924     "footwear": "footwear shoes fashion collection",
925
926     # Real Estate
927     "apartment": "apartment interior modern living room",
928     "villa": "villa luxury house exterior garden",
929     "house": "house exterior modern architecture",
930     "property": "property real estate modern building",
931     "plot": "land plot real estate",
932     "commercial space": "commercial office space interior modern",
933     "office space": "office space interior modern professional",
934     "flat": "apartment flat interior modern living",
935
936     # Education / Coaching
937     "coaching": "coaching education class students learning",
938     "online coaching": "online coaching education laptop learning",
939     "courses": "courses education learning books classroom",
940     "tuition": "tuition education students classroom learning",
941     "workshop": "workshop training education professional",
942     "classes": "classes education students learning",
943     "python": "python programming code laptop developer",
944     "java": "java programming software developer laptop",
945     "web development": "web development coding laptop developer",
946     "data science": "data science analytics laptop charts",
947     "machine learning": "machine learning artificial intelligence technology",
948     "ielts": "ielts english test preparation study books",
949     "spoken english": "english speaking class students learning",
950     "public speaking": "public speaking presentation confident stage",
951     "personality development": "personality development coaching professional confident",
952     "interview preparation": "job interview preparation coaching professional",
953     "soft skills": "soft skills training workshop professional team",
954     "leadership": "leadership training workshop professional business",
955     "career counseling": "career counseling guidance professional mentor",
956     "aptitude training": "aptitude test preparation study mathematics",
957     "gate coaching": "engineering gate exam preparation study",
958     "neet coaching": "medical neet exam preparation study books",
959     "jee coaching": "engineering jee exam preparation study",
960     "mba coaching": "mba business school preparation study professional",
961
962     # General Services
963     "consulting": "business consulting professional meeting office",
964     "photography": "photography camera professional portrait studio",
```

```
965         "videography": "videography video camera production",
966         "graphic design": "graphic design creative work computer",
967         "web design": "web design website development computer",
968         "marketing": "digital marketing strategy business creative",
969         "accounting": "accounting finance calculator business",
970         "legal": "legal law office lawyer professional",
971         "tutoring": "tutoring education student learning books",
972         "music lessons": "music lessons guitar piano teaching",
973         "art classes": "art class painting creative studio",
974         "pet grooming": "pet grooming dog salon care",
975         "veterinary": "veterinary clinic pet dog cat care",
976         "dental": "dental clinic dentist teeth care",
977         "medical": "medical clinic doctor healthcare professional",
978         "cleaning": "house cleaning service professional home",
979         "plumbing": "plumber plumbing repair tools professional",
980         "electrical": "electrician electrical work tools professional",
981         "gardening": "gardening landscaping plants flowers garden",
982         "catering": "catering food service event professional",
983         "event planning": "event planning wedding party decoration",
984         "interior design": "interior design home decor modern elegant",
985         "architecture": "architecture building design modern exterior",
986         "real estate": "real estate property house modern",
987         "bungalow": "bungalow house exterior architecture garden",
988         "penthouse": "luxury penthouse interior view modern",
989         "studio apartment": "studio apartment modern interior compact",
990         "warehouse": "industrial warehouse space interior large",
991         "shop space": "retail shop space interior modern",
992         "property management": "property management real estate professional",
993         "home loan": "home loan mortgage financial planning professional",
994         "travel": "travel destination landscape beautiful scenic",
995         "hotel": "hotel luxury room interior elegant",
996         "car rental": "car rental vehicle luxury automobile",
997         "car wash": "car wash cleaning service vehicle",
998         "repair": "repair service tools professional workshop",
999         "printing": "printing service design professional",
1000        "tailoring": "tailoring sewing clothes fashion custom",
1001        "pharmacy": "pharmacy medicine healthcare professional",
1002        "optician": "optician eyewear glasses vision care",
1003    }
1004
1005    def get_smart_query(service, btype):
1006        """Get a specific image search query based on service/product category"""
1007        svc_lower = service.lower()
1008
1009        # Try to find a specific match first
1010        for key, query in service_queries.items():
1011            if key in svc_lower:
1012                return query
1013
1014        # Business type specific fallbacks
1015        if btype in ["shop", "store", "ecommerce", "boutique"]:
1016            return f"women {service} fashion clothing"
1017        elif btype in ["salon", "spa", "beauty"]:
1018            return f"{service} beauty spa treatment"
```

```
1019     elif btype in ["gym", "fitness", "studio"]:
1020         return f"{service} fitness workout gym"
1021     elif btype in ["restaurant", "cafe", "food"]:
1022         return f"{service} food restaurant cuisine"
1023
1024     # Generic fallback
1025     return f"{service} professional high quality"
1026
1027 fallback = "https://picsum.photos/seed/business/1920/1080"
1028
1029 hero_queries = {
1030     "shop": "fashion model women clothing store",
1031     "store": "fashion model women clothing store",
1032     "ecommerce": "fashion model women clothing store",
1033     "boutique": "fashion model women clothing store",
1034     "salon": "luxury salon interior beauty spa",
1035     "spa": "luxury spa wellness relaxation interior",
1036     "beauty": "beauty salon interior elegant",
1037     "gym": "modern gym fitness equipment interior",
1038     "fitness": "fitness studio workout space",
1039     "studio": "fitness yoga studio interior",
1040     "restaurant": "elegant restaurant interior dining",
1041     "cafe": "modern cafe interior coffee shop",
1042     "food": "restaurant food interior ambiance",
1043 }
1044 about_queries = {
1045     "shop": "fashion designer workspace boutique",
1046     "store": "fashion designer workspace boutique",
1047     "ecommerce": "fashion designer workspace boutique",
1048     "boutique": "fashion designer workspace boutique",
1049     "salon": "beauty salon professional team staff",
1050     "spa": "spa therapist massage wellness professional",
1051     "beauty": "beauty professional makeup artist",
1052     "gym": "fitness trainer gym coach professional",
1053     "fitness": "personal trainer fitness professional",
1054     "studio": "fitness instructor yoga teacher",
1055     "restaurant": "chef kitchen restaurant professional",
1056     "cafe": "barista coffee cafe professional",
1057     "food": "chef cooking restaurant kitchen",
1058 }
1059 gallery_queries = {
1060     "shop": "fashion lookbook editorial style",
1061     "store": "fashion lookbook editorial style",
1062     "ecommerce": "fashion lookbook editorial style",
1063     "boutique": "fashion lookbook editorial style",
1064     "salon": "beauty salon hair makeup before after",
1065     "spa": "spa wellness relaxation massage treatment",
1066     "beauty": "beauty makeup skincare treatment",
1067     "gym": "fitness workout gym training results",
1068     "fitness": "fitness training workout motivation",
1069     "studio": "fitness yoga studio class workout",
1070     "restaurant": "restaurant food plating delicious",
1071     "cafe": "cafe coffee latte food dessert",
1072     "food": "food plating restaurant delicious meal",
```

```
1073     }
1074
1075     is_ecommerce = business_type in ["shop", "store", "ecommerce", "boutique"]
1076
1077     # Build the list of Pexels tasks (only for slots not covered by uploads)
1078     # Each task: (task_key, query, count)
1079     tasks = []
1080     if len(uploaded_urls) < 1:
1081         tasks.append(("hero", hero_queries.get(business_type.lower(), f"{business_type}
interior modern professional"), 1))
1082     if len(uploaded_urls) < 2:
1083         tasks.append(("about", about_queries.get(business_type.lower(), f"{business_type}
professional team workspace"), 1))
1084
1085     for i, service in enumerate(services[:8]):
1086         if i + 2 >= len(uploaded_urls):
1087             query = get_smart_query(service, business_type)
1088             # For ecommerce, fetch 5 images per service: [0] for card, [1:5] for product grid
1089             count = 5 if (is_ecommerce and i < 6) else 1
1090             tasks.append((f"service_{i}", query, count))
1091
1092     remaining_uploads = uploaded_urls[2 + len(services):]
1093     if not remaining_uploads:
1094         gallery_query = gallery_queries.get(business_type.lower(), f"{business_type} gallery
showcase")
1095         tasks.append(("gallery", gallery_query, 3))
1096
1097     # Run ALL Pexels fetches in parallel
1098     results = {}
1099     if tasks:
1100         print(f"[Images] Fetching {len(tasks)} image sets in parallel...")
1101         with ThreadPoolExecutor(max_workers=min(len(tasks), 12)) as executor:
1102             future_to_key = {
1103                 executor.submit(fetch_pexels_images, query, count): key
1104                 for key, query, count in tasks
1105             }
1106             for future in as_completed(future_to_key):
1107                 key = future_to_key[future]
1108                 try:
1109                     results[key] = future.result(timeout=15)
1110                 except Exception as e:
1111                     print(f"[Images] Failed for {key}: {e}")
1112                     results[key] = []
1113             print(f"[Images] All parallel fetches done.")
1114
1115     # Assign results
1116     images["hero"] = uploaded_urls[0] if uploaded_urls else (results.get("hero") or
[fallback])[0]
1117     images["about"] = uploaded_urls[1] if len(uploaded_urls) > 1 else (results.get("about")
or [fallback])[0]
1118
1119     for i, service in enumerate(services[:8]):
1120         if i + 2 < len(uploaded_urls):
1121             svc_img = uploaded_urls[i + 2]
```



```
1122         images["services"].append(svc_img)
1123         if is_ecommerce and i < 6:
1124             images["products"][service] = [svc_img] * 4
1125         else:
1126             imgs = results.get(f"service_{i}") or [fallback]
1127             images["services"].append(imgs[0])
1128             if is_ecommerce and i < 6:
1129                 # Reuse extra images fetched in the same call (no extra API call)
1130                 product_imgs = imgs[1:5] if len(imgs) > 1 else [imgs[0]] * 4
1131                 while len(product_imgs) < 4:
1132                     product_imgs.append(imgs[0])
1133                 images["products"][service] = product_imgs
1134
1135     if remaining_uploads:
1136         images["gallery"] = remaining_uploads[:6]
1137     else:
1138         images["gallery"] = results.get("gallery") or [fallback] * 3
1139
1140     return images
1141
1142
1143 # =====
1144 # WEBSITE BUILDING FROM CHAT
1145 # =====
1146
1147 def build_website_from_chat(session):
1148     """Build a complete website from chat session data"""
1149     business_type = session["business_type"]
1150     business_name = session["business_name"] or "My Business"
1151     location = session["location"] or ""
1152     description = session["description"] or ""
1153     services = session["services"] or ["Service 1", "Service 2", "Service 3"]
1154     features = session["features"] or ["contact"]
1155     style_vibe = session.get("style_vibe", "modern")
1156
1157     # Run image fetching and copywriting IN PARALLEL ? they are fully independent
1158     from concurrent.futures import ThreadPoolExecutor
1159     print("[BUILD] Fetching images + generating copy in parallel...")
1160     with ThreadPoolExecutor(max_workers=2) as executor:
1161         images_future = executor.submit(get_images_for_website, session)
1162         copy_future = executor.submit(
1163             copywriter.generate_website_copy,
1164             business_name=business_name,
1165             business_type=business_type,
1166             business_description=description,
1167             services=services,
1168             location=location,
1169             style_vibe=style_vibe,
1170         )
1171     images = images_future.result()
1172     copy_data = copy_future.result()
1173     print("[BUILD] Images + copy ready.")
1174
1175     # Build complete analysis with professional copy
```

```
1176     analysis = {
1177         "business_name": business_name,
1178         "business_type": business_type,
1179         "location": {"city": location.split(",")[0] if location else "City", "country":
location.split(",")[-1].strip() if "," in location else "Country"},
1180         "vibe": style_vibe.split()[0].lower() if style_vibe else "modern",
1181         "target_audience": "everyone",
1182         "services": services[:6],
1183         "hero_headline": copy_data.get("hero_headline", f"Welcome to {business_name}"),
1184         "hero_subtext": copy_data.get("hero_subtext", f"Experience exceptional
{business_type} services"),
1185         "tagline": copy_data.get("tagline", "Excellence in Every Detail"),
1186         "about_text": copy_data.get("about_text", f"{business_name} provides exceptional
{business_type} services."),
1187         "cta_text": copy_data.get("cta_text", "Get Started")
1188     }
1189
1190     print("[SUCCESS] Professional copy generated!")
1191
1192     # Color scheme
1193     vibe = analysis.get("vibe", "modern")
1194     color_schemes = {
1195         "luxury": {"primary": "#C9A96E", "secondary": "#1A1A1A", "accent": "#F5F5F5", "text":
"#1A1A1A"},
1196         "modern": {"primary": "#6366F1", "secondary": "#0F172A", "accent": "#F59E0B", "text":
"#1E293B"},
1197         "cozy": {"primary": "#D97706", "secondary": "#78350F", "accent": "#FEF3C7", "text":
"#451A03"},
1198         "energetic": {"primary": "#EF4444", "secondary": "#991B1B", "accent": "#FEE2E2",
"text": "#7F1D1D"},
1199         "elegant": {"primary": "#8B5CF6", "secondary": "#4C1D95", "accent": "#F3E8FF",
"text": "#581C87"},
1200         "minimalist": {"primary": "#000000", "secondary": "#374151", "accent": "#F3F4F6",
"text": "#111827"},
1201         "warm": {"primary": "#D97706", "secondary": "#78350F", "accent": "#FEF3C7", "text":
"#451A03"},
1202         "bold": {"primary": "#EF4444", "secondary": "#991B1B", "accent": "#FEE2E2", "text":
"#7F1D1D"},
1203         "clean": {"primary": "#6366F1", "secondary": "#0F172A", "accent": "#F59E0B", "text":
"#1E293B"},
1204     }
1205     colors = color_schemes.get(vibe, color_schemes["modern"])
1206
1207     # Generate HTML with functional features
1208     html = generate_full_website(analysis, colors, images, features, session)
1209
1210     # Save
1211     website_session_id = session.get("website_session_id") or str(uuid.uuid4())
1212     html_path = f"generated_websites/{website_session_id}.html"
1213     state_path = f"website_states/{website_session_id}.json"
1214
1215     with open(html_path, "w", encoding="utf-8") as f:
1216         f.write(html)
1217
```

```
1218     state = {
1219         "version": 1,
1220         "created_at": datetime.now().isoformat(),
1221         "business_description": description,
1222         "analysis": analysis,
1223         "colors": colors,
1224         "images": images,
1225         "features": features,
1226         "chat_session": session.get("conversation_history", [])[-5:], # Last 5 messages
1227     }
1228
1229     with open(state_path, "w", encoding="utf-8") as f:
1230         json.dump(state, f, indent=2)
1231
1232     WEBSITE_STATES[website_session_id] = state
1233
1234     return {"session_id": website_session_id, "html": html, "analysis": analysis}
1235
1236
1237 def edit_website_from_chat(session, edit_request):
1238     """Edit an existing website from chat"""
1239     website_session_id = session["website_session_id"]
1240     if not website_session_id:
1241         raise Exception("No website to edit")
1242
1243     state_path = f"website_states/{website_session_id}.json"
1244     if not os.path.exists(state_path):
1245         raise Exception("Website state not found")
1246
1247     with open(state_path, "r", encoding="utf-8") as f:
1248         state = json.load(f)
1249
1250     original_desc = state.get("business_description", "")
1251     analysis = state.get("analysis", {})
1252     features = state.get("features", session.get("features", ["contact"]))
1253
1254     # Create modified description
1255     modified_desc = (
1256         f"{original_desc} (Business: {analysis.get('business_name', '')}), "
1257         f"Type: {analysis.get('business_type', '')}). "
1258         f"Apply these changes: {edit_request}"
1259     )
1260
1261     # Rebuild
1262     session["description"] = modified_desc
1263     result = build_website_from_chat(session)
1264
1265     # Update version
1266     new_version = state.get("version", 1) + 1
1267     updated_state = WEBSITE_STATES.get(result["session_id"], {})
1268     updated_state["version"] = new_version
1269     WEBSITE_STATES[result["session_id"]] = updated_state
1270
1271     state_path = f"website_states/{result['session_id']}.json"
```

--- Middle Section Omitted ---

As per the Indian Copyright Office requirement for computer programs exceeding 20 pages, only the first 25 pages and last 25 pages of the source code are included in this submission. The complete source code is available for inspection upon request.

```
157     background: var(--bg-input);
158     border: 1.5px solid var(--border);
159     border-radius: 13px;
160     display: flex;
161     flex-direction: column;
162     gap: 10px;
163     transition: background 0.3s ease, border-color 0.3s ease;
164 }
165 .result-deploy-label {
166     font-size: 12px;
167     font-weight: 700;
168     color: var(--text-muted);
169     text-transform: uppercase;
170     letter-spacing: 0.5px;
171 }
172
173 /* Right panel ? browser frame */
174 .result-right {
175     display: flex;
176     flex-direction: column;
177     background: var(--bg-page);
178     overflow: hidden;
179     transition: background 0.3s ease;
180 }
181 .result-preview-bar {
182     display: flex; align-items: center; gap: 6px;
183     padding: 10px 16px;
184     background: var(--bg-card);
185     border-bottom: 1px solid var(--border);
186     flex-shrink: 0;
187     transition: background 0.3s ease, border-color 0.3s ease;
188 }
189 .result-preview-dot { width: 12px; height: 12px; border-radius: 50%; }
190 .result-preview-dot.red { background: #ef4444; }
191 .result-preview-dot.yellow { background: #f59e0b; }
192 .result-preview-dot.green { background: #10b981; }
193 .result-preview-url {
194     flex: 1; margin-left: 10px;
195     background: var(--bg-input);
196     border: 1px solid var(--border);
197     border-radius: 6px; padding: 5px 12px;
198     font-size: 12px; color: var(--text-muted); font-family: monospace;
199     transition: background 0.3s ease, border-color 0.3s ease;
200 }
201 .result-iframe { flex: 1; width: 100%; border: none; background: #fff; }
202
203 @media (max-width: 768px) {
204     .result-body { grid-template-columns: 1fr; height: auto; }
205     .result-right { display: none; }
206 }
207
208 /* Fullscreen preview modal ? appears on mobile when Open Preview is tapped */
209 .result-preview-modal {
210     position: fixed;
```

```
211     inset: 0;
212     z-index: 1000;
213     display: flex;
214     flex-direction: column;
215     background: #fff;
216 }
217 .result-preview-modal-bar {
218     display: flex;
219     align-items: center;
220     padding: 10px 14px;
221     background: var(--bg-card);
222     border-bottom: 1px solid var(--border);
223     flex-shrink: 0;
224     gap: 8px;
225 }
226 .result-preview-modal-close {
227     flex-shrink: 0;
228     width: 34px;
229     height: 34px;
230     border-radius: 8px;
231     border: 1.5px solid var(--border);
232     background: var(--bg-subtle);
233     color: var(--text-secondary);
234     font-size: 15px;
235     font-weight: 700;
236     cursor: pointer;
237     display: flex;
238     align-items: center;
239     justify-content: center;
240     transition: all 0.15s;
241 }
242 .result-preview-modal-close:hover {
243     border-color: #ef4444;
244     color: #ef4444;
245     background: #fef2f2;
246 }
247
```

File 23: frontend/src/Onboarding.css

Frontend - Onboarding.css

```
1  /* ????????????????????????????????????????????????????????????? */
2      Onboarding ? Sitekraft AI wizard · Light + Moonlight dark
3      ????????????????????????????????????????????????????????????? */
4
5  /* ?? Root ??????????????????????????????????????????????????????? */
6  .ob-root {
7      min-height: 100vh;
8      background: var(--bg-page);
9      display: flex;
10     flex-direction: column;
11     font-family: -apple-system, BlinkMacSystemFont, 'Segoe UI', 'Inter', Roboto, sans-serif;
12     color: var(--text-primary);
13     transition: background 0.3s ease, color 0.3s ease;
14 }
15
16 /* ?? Moonlight star background (dark only) ?????????????????????? */
17 [data-theme="dark"] .ob-root::before {
18     content: '';
19     position: fixed;
20     inset: 0;
21     pointer-events: none;
22     background:
23         radial-gradient(ellipse 80% 50% at 20% 10%, rgba(139,92,246,0.12) 0%, transparent 60%),
24         radial-gradient(ellipse 60% 40% at 80% 80%, rgba(99,102,241,0.08) 0%, transparent 60%);
25     z-index: 0;
26 }
27 [data-theme="dark"] .ob-header,
28 [data-theme="dark"] .ob-progress-track,
29 [data-theme="dark"] .ob-main {
30     position: relative;
31     z-index: 1;
32 }
33
34 /* ?? Header ?????????????????????????????????????????????????????? */
35 .ob-header {
36     display: flex;
37     justify-content: space-between;
38     align-items: center;
39     padding: 14px 28px;
40     background: var(--header-bg);
41     backdrop-filter: blur(12px);
42     -webkit-backdrop-filter: blur(12px);
43     border-bottom: 1px solid var(--border);
44     position: sticky;
45     top: 0;
46     z-index: 100;
47     transition: background 0.3s ease, border-color 0.3s ease;
48 }
49 .ob-brand {
50     font-size: 18px;
51     font-weight: 800;
```

```
52     background: linear-gradient(135deg, var(--accent), var(--accent-2));
53     -webkit-background-clip: text;
54     -webkit-text-fill-color: transparent;
55     background-clip: text;
56     letter-spacing: -0.5px;
57 }
58 .ob-header-right {
59     display: flex;
60     align-items: center;
61     gap: 14px;
62 }
63 .ob-step-counter {
64     font-size: 13px;
65     color: var(--text-muted);
66     font-weight: 500;
67 }
68
69 /* ?? Theme toggle button ?????????????????????????????????????? */
70 .ob-theme-btn {
71     width: 36px;
72     height: 36px;
73     border-radius: 10px;
74     border: 1.5px solid var(--border);
75     background: var(--bg-subtle);
76     color: var(--text-secondary);
77     font-size: 16px;
78     cursor: pointer;
79     display: flex;
80     align-items: center;
81     justify-content: center;
82     transition: all 0.2s ease;
83     flex-shrink: 0;
84 }
85 .ob-theme-btn:hover {
86     border-color: var(--accent);
87     color: var(--accent);
88     background: var(--accent-light);
89     transform: rotate(15deg);
90 }
91
92 /* ?? Progress bar ?????????????????????????????????????????????? */
93 .ob-progress-track {
94     height: 3px;
95     background: var(--progress-bg);
96     transition: background 0.3s ease;
97 }
98 .ob-progress-fill {
99     height: 100%;
100     background: linear-gradient(90deg, var(--accent), var(--accent-2));
101     transition: width 0.45s cubic-bezier(0.4, 0, 0.2, 1);
102     box-shadow: 0 0 8px var(--glow-strong);
103 }
104
105 /* ?? Main / Card ?????????????????????????????????????????????? */
```



```
106 .ob-main {
107     flex: 1;
108     display: flex;
109     justify-content: center;
110     align-items: flex-start;
111     padding: 48px 20px 80px;
112 }
113 .ob-card {
114     background: var(--bg-card);
115     border-radius: 22px;
116     padding: 40px 44px;
117     width: 100%;
118     max-width: 540px;
119     box-shadow: var(--shadow-card);
120     border: 1px solid var(--border);
121     animation: ob-fadein 0.28s ease;
122     transition: background 0.3s ease, border-color 0.3s ease, box-shadow 0.3s ease;
123 }
124 [data-theme="dark"] .ob-card {
125     box-shadow: var(--shadow-card), 0 0 0 1px rgba(139,92,246,0.08);
126 }
127 @keyframes ob-fadein {
128     from { opacity: 0; transform: translateY(12px); }
129     to   { opacity: 1; transform: translateY(0); }
130 }
131
132 /* ?? Step inner layout ?????????????????????????????????????????????? */
133 .ob-step {
134     display: flex;
135     flex-direction: column;
136     gap: 18px;
137 }
138 .ob-step-head { margin-bottom: 2px; }
139 .ob-title {
140     font-size: 22px;
141     font-weight: 700;
142     color: var(--text-primary);
143     margin: 0 0 8px;
144     line-height: 1.3;
145 }
146 .ob-subtitle {
147     font-size: 14px;
148     color: var(--text-secondary);
149     margin: 0;
150     line-height: 1.55;
151 }
152 .ob-back {
153     display: inline-flex;
154     align-items: center;
155     gap: 4px;
156     background: none;
157     border: none;
158     color: var(--text-muted);
159     font-size: 13px;
```

```
160     font-weight: 500;
161     cursor: pointer;
162     padding: 0;
163     transition: color 0.15s;
164 }
165 .ob-back:hover { color: var(--accent); }
166
167 /* ?? Inputs ????????????????????????????????????????????????????????? */
168 .ob-input {
169     width: 100%;
170     padding: 12px 14px;
171     border: 1.5px solid var(--border);
172     border-radius: 11px;
173     font-size: 15px;
174     color: var(--text-primary);
175     background: var(--bg-input);
176     outline: none;
177     transition: border-color 0.18s, box-shadow 0.18s, background 0.18s;
178     box-sizing: border-box;
179     font-family: inherit;
180 }
181 .ob-input:focus {
182     border-color: var(--border-focus);
183     background: var(--bg-input-focus);
184     box-shadow: 0 0 0 3px var(--glow);
185 }
186 .ob-input::placeholder { color: var(--text-muted); }
187 .ob-input-xl {
188     font-size: 18px;
189     padding: 15px 18px;
190 }
191 .ob-textarea {
192     width: 100%;
193     padding: 11px 14px;
194     border: 1.5px solid var(--border);
195     border-radius: 11px;
196     font-size: 14px;
197     color: var(--text-primary);
198     background: var(--bg-input);
199     outline: none;
200     resize: vertical;
201     font-family: inherit;
202     transition: border-color 0.18s, box-shadow 0.18s, background 0.18s;
203     box-sizing: border-box;
204 }
205 .ob-textarea:focus {
206     border-color: var(--border-focus);
207     background: var(--bg-input-focus);
208     box-shadow: 0 0 0 3px var(--glow);
209 }
210
211 /* ?? Primary nav button ????????????????????????????????????????????? */
212 .ob-primary-btn {
213     width: 100%;
```

```
214     padding: 14px;
215     border-radius: 12px;
216     background: linear-gradient(135deg, var(--accent), var(--accent-2));
217     color: #fff;
218     font-size: 15px;
219     font-weight: 600;
220     border: none;
221     cursor: pointer;
222     transition: transform 0.18s, box-shadow 0.18s, opacity 0.18s;
223     font-family: inherit;
224     position: relative;
225     overflow: hidden;
226 }
227 .ob-primary-btn::after {
228     content: '';
229     position: absolute;
230     inset: 0;
231     background: linear-gradient(135deg, rgba(255,255,255,0.1), transparent);
232     opacity: 0;
233     transition: opacity 0.18s;
234 }
235 .ob-primary-btn:hover:not(:disabled)::after { opacity: 1; }
236 .ob-primary-btn:hover:not(:disabled) {
237     transform: translateY(-1px);
238     box-shadow: var(--shadow-btn);
239 }
240 .ob-primary-btn:disabled {
241     background: var(--border);
242     color: var(--text-muted);
243     cursor: not-allowed;
244     transform: none;
245     box-shadow: none;
246 }
247
248 /* ?? Step 2 · Business type grid ?????????????????????????????? */
249 .ob-type-grid {
250     display: grid;
251     grid-template-columns: 1fr 1fr;
252     gap: 10px;
253 }
254 .ob-type-card {
255     display: flex;
256     flex-direction: column;
257     align-items: center;
258     gap: 8px;
259     padding: 18px 10px;
260     border: 1.5px solid var(--border);
261     border-radius: 14px;
262     background: var(--bg-input);
263     cursor: pointer;
264     font-family: inherit;
265     transition: border-color 0.17s, background 0.17s, transform 0.17s, box-shadow 0.17s;
266 }
267 .ob-type-card:hover {
```

```
268     border-color: var(--accent);
269     background: var(--accent-light);
270     transform: translateY(-2px);
271     box-shadow: 0 4px 16px var(--glow);
272 }
273 .ob-type-emoji { font-size: 26px; }
274 .ob-type-label {
275     font-size: 13px;
276     font-weight: 600;
277     color: var(--text-secondary);
278     text-align: center;
279     line-height: 1.3;
280 }
281
282 /* Step 2 · selling-type confirmation */
283 .ob-confirm-list {
284     display: flex;
285     flex-direction: column;
286     gap: 9px;
287 }
288 .ob-confirm-yes {
289     padding: 14px 18px;
290     border-radius: 12px;
291     background: linear-gradient(135deg, var(--accent), var(--accent-2));
292     color: #fff;
293     font-size: 15px;
294     font-weight: 600;
295     border: none;
296     cursor: pointer;
297     font-family: inherit;
298     text-align: left;
299     transition: opacity 0.15s, box-shadow 0.15s;
300 }
301 .ob-confirm-yes:hover { opacity: 0.9; box-shadow: var(--shadow-btn); }
302 .ob-confirm-alt {
303     padding: 13px 18px;
304     border: 1.5px solid var(--border);
305     border-radius: 12px;
306     background: var(--bg-input);
307     font-size: 14px;
308     font-weight: 500;
309     color: var(--text-primary);
310     cursor: pointer;
311     text-align: left;
312     font-family: inherit;
313     transition: border-color 0.15s, background 0.15s;
314 }
315 .ob-confirm-alt:hover {
316     border-color: var(--accent);
317     background: var(--accent-light);
318 }
319
320 /* ?? Step 4 · Style selector ?????????????????????????????????? */
321 .ob-style-list {
```

```
322     display: flex;
323     flex-direction: column;
324     gap: 9px;
325 }
326 .ob-style-card {
327     display: flex;
328     align-items: center;
329     gap: 14px;
330     padding: 13px 14px;
331     border: 1.5px solid var(--border);
332     border-radius: 13px;
333     background: var(--bg-input);
334     cursor: pointer;
335     font-family: inherit;
336     text-align: left;
337     transition: border-color 0.17s, background 0.17s, box-shadow 0.17s;
338 }
339 .ob-style-card:hover, .ob-style-selected {
340     border-color: var(--accent);
341     background: var(--accent-light);
342 }
343 .ob-style-selected {
344     box-shadow: 0 0 0 2px var(--glow-strong);
345 }
346 .ob-style-swatch {
347     width: 50px;
348     height: 38px;
349     border-radius: 8px;
350     flex-shrink: 0;
351     display: flex;
352     align-items: center;
353     justify-content: center;
354     border: 1px solid rgba(0,0,0,0.08);
355 }
356 .ob-style-text { display: flex; flex-direction: column; gap: 2px; }
357 .ob-style-name { font-size: 14px; font-weight: 600; color: var(--text-primary); }
358 .ob-style-desc { font-size: 12px; color: var(--text-secondary); }
359
360 /* ?? Step 5 · Color picker ?????????????????????????????????????? */
361 .ob-color-wrap {
362     display: flex;
363     flex-direction: column;
364     gap: 14px;
365 }
366 .ob-color-picker-row {
367     display: flex;
368     align-items: center;
369     gap: 12px;
370 }
371 .ob-color-input {
372     width: 54px;
373     height: 54px;
374     border: none;
375     border-radius: 12px;
```

```
376     cursor: pointer;
377     padding: 3px;
378     background: none;
379 }
380 .ob-color-hex {
381     font-size: 16px;
382     font-weight: 600;
383     color: var(--text-primary);
384     font-family: 'SF Mono', 'Fira Code', monospace;
385 }
386 .ob-quick-colors {
387     display: flex;
388     gap: 9px;
389     flex-wrap: wrap;
390 }
391 .ob-dot {
392     width: 32px;
393     height: 32px;
394     border-radius: 50%;
395     border: 2.5px solid transparent;
396     cursor: pointer;
397     transition: transform 0.15s, border-color 0.15s;
398 }
399 .ob-dot:hover, .ob-dot-active {
400     border-color: var(--text-primary);
401     transform: scale(1.18);
402 }
403 .ob-ai-color-btn {
404     padding: 12px 18px;
405     border: 1.5px dashed var(--border-accent);
406     border-radius: 12px;
407     background: var(--accent-light);
408     color: var(--accent);
409     font-size: 14px;
410     font-weight: 600;
411     cursor: pointer;
412     font-family: inherit;
413     transition: all 0.18s;
414 }
415 .ob-ai-color-btn:hover, .ob-ai-active {
416     background: linear-gradient(135deg, var(--accent), var(--accent-2));
417     color: #fff;
418     border-color: transparent;
419     box-shadow: var(--shadow-btn);
420 }
421
422 /* ?? Step 6 · Products / Services ?????????????????????????????? */
423 .ob-items-scroll {
424     display: flex;
425     flex-direction: column;
426     gap: 12px;
427     max-height: 360px;
428     overflow-y: auto;
429     padding-right: 4px;
```

```
430 }
431 .ob-items-scroll::-webkit-scrollbar { width: 4px; }
432 .ob-items-scroll::-webkit-scrollbar-track { background: transparent; }
433 .ob-items-scroll::-webkit-scrollbar-thumb {
434     background: var(--border-accent);
435     border-radius: 4px;
436 }
437
438 .ob-item-card {
439     display: flex;
440     flex-direction: column;
441     gap: 9px;
442     padding: 14px;
443     border: 1.5px solid var(--border);
444     border-radius: 13px;
445     background: var(--bg-input);
446     transition: border-color 0.2s, box-shadow 0.2s;
447 }
448 .ob-item-card:focus-within {
449     border-color: var(--border-focus);
450     box-shadow: 0 0 0 3px var(--glow);
451 }
452 .ob-item-row-head {
453     display: flex;
454     justify-content: space-between;
455     align-items: center;
456 }
457 .ob-item-badge {
458     font-size: 11px;
459     font-weight: 700;
460     color: var(--accent);
461     text-transform: uppercase;
462     letter-spacing: 0.6px;
463 }
464 .ob-remove {
465     background: none;
466     border: none;
467     color: var(--text-muted);
468     cursor: pointer;
469     font-size: 14px;
470     padding: 2px 6px;
471     border-radius: 6px;
472     transition: background 0.15s, color 0.15s;
473 }
474 .ob-remove:hover { background: var(--danger-bg); color: var(--danger); }
475 .ob-add-btn {
476     width: 100%;
477     padding: 11px;
478     border: 1.5px dashed var(--border-accent);
479     border-radius: 12px;
480     background: var(--accent-light);
481     color: var(--accent);
482     font-size: 14px;
483     font-weight: 600;
```

```
484     cursor: pointer;
485     font-family: inherit;
486     transition: background 0.15s, border-color 0.15s;
487 }
488 .ob-add-btn:hover {
489     background: var(--bg-subtle);
490     border-color: var(--accent);
491 }
492
493 /* ?? Step 6 · Per-item image upload ?????????????????????????? */
494 .ob-item-img-row { margin-top: 2px; }
495 .ob-item-upload-btn {
496     display: inline-flex;
497     align-items: center;
498     gap: 6px;
499     padding: 8px 14px;
500     border: 1.5px dashed var(--border-accent);
501     border-radius: 9px;
502     background: var(--accent-light);
503     color: var(--accent);
504     font-size: 13px;
505     font-weight: 600;
506     cursor: pointer;
507     transition: background 0.15s, border-color 0.15s;
508 }
509 .ob-item-upload-btn:hover {
510     background: var(--bg-subtle);
511     border-color: var(--accent);
512 }
513 .ob-item-thumb-wrap {
514     position: relative;
515     display: inline-block;
516 }
517 .ob-item-thumb {
518     width: 80px;
519     height: 60px;
520     object-fit: cover;
521     border-radius: 8px;
522     border: 1.5px solid var(--border);
523     display: block;
524 }
525 .ob-item-uploaded-badge {
526     display: block;
527     font-size: 11px;
528     font-weight: 700;
529     color: var(--success-color);
530     text-align: center;
531     margin-top: 4px;
532     letter-spacing: 0.01em;
533 }
534 .ob-item-thumb-remove {
535     position: absolute;
536     top: -6px;
537     right: -6px;
```



```

538     width: 20px;
539     height: 20px;
540     border-radius: 50%;
541     background: var(--danger);
542     color: #fff;
543     border: none;
544     font-size: 10px;
545     font-weight: 700;
546     cursor: pointer;
547     display: flex;
548     align-items: center;
549     justify-content: center;
550     line-height: 1;
551 }
552
553 /* ?? Step 7 · Photos ????????????????????????????????????????????????????????????? */
554 .ob-choice-list {
555     display: flex;
556     flex-direction: column;
557     gap: 10px;
558 }
559 .ob-choice-card {
560     display: flex;
561     align-items: center;
562     gap: 16px;
563     padding: 18px;
564     border: 1.5px solid var(--border);
565     border-radius: 14px;
566     background: var(--bg-input);
567     cursor: pointer;
568     font-family: inherit;
569     text-align: left;
570     transition: border-color 0.17s, background 0.17s, transform 0.17s, box-shadow 0.17s;
571 }
572 .ob-choice-card:hover {
573     border-color: var(--accent);
574     background: var(--accent-light);
575     transform: translateY(-1px);
576     box-shadow: 0 4px 16px var(--glow);
577 }
578 .ob-choice-icon { font-size: 26px; flex-shrink: 0; }
579 .ob-choice-text { display: flex; flex-direction: column; gap: 3px; }
580 .ob-choice-title { font-size: 15px; font-weight: 600; color: var(--text-primary); }
581 .ob-choice-sub { font-size: 13px; color: var(--text-secondary); }
582
583 .ob-upload-list { display: flex; flex-direction: column; gap: 10px; }
584 .ob-upload-row {
585     display: flex;
586     justify-content: space-between;
587     align-items: center;
588     padding: 13px 15px;
589     border: 1.5px solid var(--border);
590     border-radius: 12px;
591     background: var(--bg-input);

```

```
592     gap: 12px;
593   }
594   .ob-upload-info { display: flex; flex-direction: column; gap: 2px; }
595   .ob-upload-label { font-size: 14px; font-weight: 600; color: var(--text-primary); }
596   .ob-upload-hint { font-size: 12px; color: var(--text-muted); }
597   .ob-upload-btn {
598     padding: 8px 14px;
599     border-radius: 9px;
600     background: var(--accent-light);
601     color: var(--accent);
602     font-size: 13px;
603     font-weight: 600;
604     cursor: pointer;
605     border: 1.5px solid var(--border-accent);
606     white-space: nowrap;
607     transition: background 0.15s, color 0.15s, border-color 0.15s;
608     flex-shrink: 0;
609   }
610   .ob-upload-btn:hover { background: var(--accent); color: #fff; border-color: var(--accent); }
611   .ob-uploaded { background: var(--success-bg); color: var(--success-color); border-color:
var(--success-border); }
612   .ob-uploaded:hover { background: var(--success-color); color: #fff; border-color:
var(--success-color); }
613
614   /* ?? Step 8 · Contact ????????????????????????????????????????????????????????????? */
615   .ob-contact-grid { display: flex; flex-direction: column; gap: 12px; }
616   .ob-contact-field { display: flex; flex-direction: column; gap: 5px; }
617   .ob-field-label { font-size: 13px; font-weight: 600; color: var(--text-secondary); }
618   .ob-required { color: var(--danger); }
619   .ob-opt { font-weight: 400; color: var(--text-muted); }
620
621   /* ?? Submit button ????????????????????????????????????????????????????????????? */
622   .ob-submit-btn {
623     width: 100%;
624     padding: 15px;
625     border-radius: 13px;
626     background: linear-gradient(135deg, var(--accent), var(--accent-2));
627     color: #fff;
628     font-size: 16px;
629     font-weight: 700;
630     border: none;
631     cursor: pointer;
632     display: flex;
633     align-items: center;
634     justify-content: center;
635     gap: 10px;
636     font-family: inherit;
637     transition: transform 0.18s, box-shadow 0.18s, opacity 0.18s;
638     margin-top: 4px;
639     position: relative;
640     overflow: hidden;
641   }
642   .ob-submit-btn::after {
643     content: '';
```

```
644     position: absolute;
645     inset: 0;
646     background: linear-gradient(135deg, rgba(255,255,255,0.12), transparent);
647     opacity: 0;
648     transition: opacity 0.18s;
649 }
650 .ob-submit-btn:hover:not(:disabled)::after { opacity: 1; }
651 .ob-submit-btn:hover:not(:disabled) {
652     transform: translateY(-2px);
653     box-shadow: var(--shadow-btn);
654 }
655 .ob-submit-btn:disabled {
656     opacity: 0.5;
657     cursor: not-allowed;
658     transform: none;
659 }
660 .ob-submitting { opacity: 0.85; }
661
662 /* Spinner */
663 .ob-spinner {
664     width: 17px;
665     height: 17px;
666     border: 2.5px solid rgba(255,255,255,0.35);
667     border-top-color: #fff;
668     border-radius: 50%;
669     display: inline-block;
670     animation: ob-spin 0.65s linear infinite;
671 }
672 @keyframes ob-spin { to { transform: rotate(360deg); } }
673
674 /* ?? Mobile enhancements (??768px) ?????????????????????????? */
675 @media (max-width: 768px) {
676
677     /* Subtle top + bottom blobs ? lighter than desktop so they don't overpower content */
678     .ob-root {
679         background:
680             radial-gradient(ellipse 80% 35% at 50% 0%,    rgba(99,102,241,0.10) 0%, transparent
681 70%),
682             radial-gradient(ellipse 70% 30% at 50% 100%,   rgba(139,92,246,0.08) 0%, transparent
683 70%),
684             var(--bg-page);
685     }
686     [data-theme="dark"] .ob-root {
687         background:
688             radial-gradient(ellipse 80% 35% at 50% 0%,    rgba(139,92,246,0.22) 0%, transparent
689 70%),
690             radial-gradient(ellipse 70% 30% at 50% 100%,   rgba(99,102,241,0.14) 0%, transparent
691 70%),
692             var(--bg-page);
693     }
694
695     /* Card fills width nicely with a soft shadow */
696     .ob-card {
697         border-radius: 20px;
```

```
694     box-shadow:
695         0 2px 8px rgba(0,0,0,0.06),
696         0 8px 32px rgba(99,102,241,0.10);
697 }
698 [data-theme="dark"] .ob-card {
699     box-shadow:
700         0 2px 8px rgba(0,0,0,0.4),
701         0 8px 32px rgba(139,92,246,0.18),
702         0 0 0 1px rgba(139,92,246,0.10);
703 }
704
705 /* Compact but polished header */
706 .ob-header {
707     padding: 12px 18px;
708 }
709 .ob-brand {
710     font-size: 17px;
711 }
712 .ob-theme-btn {
713     width: 34px;
714     height: 34px;
715     font-size: 15px;
716 }
717
718 /* Tighter main padding */
719 .ob-main {
720     padding: 24px 14px 56px;
721     align-items: flex-start;
722 }
723
724 /* Slightly smaller title so it fits on one line */
725 .ob-title { font-size: 19px; }
726 .ob-card { padding: 26px 20px; }
727
728 /* Single column business type grid */
729 .ob-type-grid { grid-template-columns: 1fr 1fr; }
730
731 /* Progress bar a tiny bit thicker on mobile ? easier to see */
732 .ob-progress-track { height: 4px; }
733
734 /* Input font comfortable on mobile ? prevents iOS auto-zoom */
735 .ob-input { font-size: 16px; }
736 .ob-textarea { font-size: 16px; }
737 }
738
739 /* ?? Desktop enhancements (?769px) ? mobile completely unchanged ?? */
740 @media (min-width: 769px) {
741
742     /* Rich gradient background so the page never looks empty */
743     .ob-root {
744         background:
745             radial-gradient(ellipse 55% 60% at 8% 15%,    rgba(99,102,241,0.09) 0%, transparent
746             65%),
747             radial-gradient(ellipse 45% 55% at 92% 85%,    rgba(139,92,246,0.07) 0%, transparent
```

```
        65%),
747      radial-gradient(ellipse 35% 40% at 50% 50%, rgba(168,85,247,0.04) 0%, transparent
55%),
748      var(--bg-page);
749    }
750
751    /* Dark mode gets deeper, more dramatic blobs */
752    [data-theme="dark"] .ob-root {
753      background:
754        radial-gradient(ellipse 55% 60% at 8% 15%, rgba(139,92,246,0.18) 0%, transparent 65%),
755        radial-gradient(ellipse 45% 55% at 92% 85%, rgba(99,102,241,0.12) 0%, transparent
65%),
756        radial-gradient(ellipse 30% 35% at 60% 30%, rgba(167,139,250,0.08) 0%, transparent
50%),
757        var(--bg-page);
758    }
759
760    /* Decorative floating orbs ? purely visual, pointer-events off */
761    .ob-main {
762      position: relative;
763      align-items: center;
764      padding-top: 40px;
765      padding-bottom: 60px;
766    }
767    .ob-main::before,
768    .ob-main::after {
769      content: '';
770      position: fixed;
771      border-radius: 50%;
772      pointer-events: none;
773      z-index: 0;
774      animation: ob-float 8s ease-in-out infinite;
775    }
776    .ob-main::before {
777      width: 340px; height: 340px;
778      top: 8%; left: -80px;
779      background: radial-gradient(circle, rgba(99,102,241,0.10) 0%, transparent 70%);
780      animation-delay: 0s;
781    }
782    .ob-main::after {
783      width: 280px; height: 280px;
784      bottom: 10%; right: -60px;
785      background: radial-gradient(circle, rgba(139,92,246,0.09) 0%, transparent 70%);
786      animation-delay: -4s;
787    }
788    [data-theme="dark"] .ob-main::before {
789      background: radial-gradient(circle, rgba(139,92,246,0.20) 0%, transparent 70%);
790    }
791    [data-theme="dark"] .ob-main::after {
792      background: radial-gradient(circle, rgba(99,102,241,0.16) 0%, transparent 70%);
793    }
794    @keyframes ob-float {
795      0%, 100% { transform: translateY(0px) scale(1); }
796      50%      { transform: translateY(-24px) scale(1.04); }
```

```
797     }
798
799     /* Card sits above the orbs */
800     .ob-card {
801         position: relative;
802         z-index: 1;
803         max-width: 560px;
804         box-shadow:
805             var(--shadow-card),
806             0 20px 60px rgba(99,102,241,0.08);
807     }
808     [data-theme="dark"] .ob-card {
809         box-shadow:
810             var(--shadow-card),
811             0 20px 60px rgba(139,92,246,0.15),
812             0 0 0 1px rgba(139,92,246,0.10);
813     }
814
815     /* Bigger, more visible brand name */
816     .ob-brand { font-size: 21px; }
817
818     /* Bigger theme toggle so it's easy to spot */
819     .ob-theme-btn {
820         width: 40px;
821         height: 40px;
822         font-size: 18px;
823         border-radius: 12px;
824     }
825 }
826
```

File 24: frontend/src/index.cssFrontend - index.css

```
1  /* ?? Design tokens ? Light (default) ?????????????????????? */
2  :root {
3      --bg-page:          #f1f5f9;
4      --bg-card:          #ffffff;
5      --bg-input:         #f8fafc;
6      --bg-input-focus:   #ffffff;
7      --bg-subtle:        #eef2ff;
8      --border:           #e2e8f0;
9      --border-focus:     #6366f1;
10     --border-accent:     #c7d2fe;
11     --text-primary:      #0f172a;
12     --text-secondary:    #64748b;
13     --text-muted:        #94a3b8;
14     --accent:            #6366f1;
15     --accent-2:          #8b5cf6;
16     --accent-light:      #eef2ff;
17     --glow:              rgba(99,102,241,0.12);
18     --glow-strong:       rgba(99,102,241,0.28);
19     --shadow-card:       0 1px 3px rgba(0,0,0,0.06), 0 4px 16px rgba(0,0,0,0.07);
20     --shadow-btn:        0 6px 18px rgba(99,102,241,0.35);
21     --header-bg:         rgba(255,255,255,0.92);
22     --progress-bg:       #e2e8f0;
23     --success-bg:        #d1fae5;
24     --success-color:     #059669;
25     --success-border:    #6ee7b7;
26     --danger:            #ef4444;
27     --danger-bg:         #fee2e2;
28 }
29
30 /* ?? Moonlight Dark theme ?????????????????????? */
31 [data-theme="dark"] {
32     --bg-page:          #0d1117;
33     --bg-card:          #161b2e;
34     --bg-input:         #1e2640;
35     --bg-input-focus:   #232d4a;
36     --bg-subtle:        #1a2035;
37     --border:           #2a3550;
38     --border-focus:     #a78bfa;
39     --border-accent:    #4c3d8f;
40     --text-primary:     #e2e8f0;
41     --text-secondary:   #8892a4;
42     --text-muted:       #5a6580;
43     --accent:           #8b5cf6;
44     --accent-2:         #a78bfa;
45     --accent-light:     #1e1a3a;
46     --glow:             rgba(139,92,246,0.15);
47     --glow-strong:      rgba(139,92,246,0.35);
48     --shadow-card:      0 1px 3px rgba(0,0,0,0.4), 0 4px 24px rgba(0,0,0,0.5);
49     --shadow-btn:       0 6px 24px rgba(139,92,246,0.45);
50     --header-bg:        rgba(13,17,23,0.92);
51     --progress-bg:      #1e2640;
```

```
52     --success-bg:      #0d2a1f;
53     --success-color:   #34d399;
54     --success-border:  #065f46;
55     --danger:          #f87171;
56     --danger-bg:       #2a1515;
57 }
58
59 *, ::before, ::after { box-sizing: border-box; margin: 0; padding: 0; }
60
61 body {
62     min-height: 100vh;
63     font-family: -apple-system, BlinkMacSystemFont, 'Segoe UI', 'Inter', Roboto, sans-serif;
64     -webkit-font-smoothing: antialiased;
65     background: var(--bg-page);
66     color: var(--text-primary);
67     transition: background 0.3s ease, color 0.3s ease;
68 }
69
70 #root { min-height: 100vh; }
71
```


File 25: frontend/index.html

Frontend - index.html

```
1    <!doctype html>
2    <html lang="en">
3      <head>
4        <meta charset="UTF-8" />
5        <link rel="icon" type="image/svg+xml" href="/vite.svg" />
6        <meta name="viewport" content="width=device-width, initial-scale=1.0" />
7        <title>Sitekraft</title>
8      </head>
9      <body>
10       <div id="root"></div>
11       <script type="module" src="/src/main.jsx"></script>
12     </body>
13   </html>
14
```

File 26: frontend/vite.config.js

Frontend - vite.config.js

```
1  import { defineConfig } from 'vite'
2  import react from '@vitejs/plugin-react'
3
4  // https://vite.dev/config/
5  export default defineConfig({
6    plugins: [react()],
7    server: {
8      proxy: {
9        '/api': {
10          target: 'http://localhost:8000',
11          changeOrigin: true,
12        }
13      }
14    }
15  })
16
```

File 27: frontend/eslint.config.js

Frontend - eslint.config.js

```
1   import js from '@eslint/js'
2   import globals from 'globals'
3   import reactHooks from 'eslint-plugin-react-hooks'
4   import reactRefresh from 'eslint-plugin-react-refresh'
5   import { defineConfig, globalIgnores } from 'eslint/config'
6
7   export default defineConfig([
8     globalIgnores(['dist']),
9     {
10      files: ['**/*.js', '**/*.jsx'],
11      extends: [
12        js.configs.recommended,
13        reactHooks.configs.flat.recommended,
14        reactRefresh.configs.vite,
15      ],
16      languageOptions: {
17        ecmaVersion: 2020,
18        globals: globals.browser,
19        parserOptions: {
20          ecmaVersion: 'latest',
21          ecmaFeatures: { jsx: true },
22          sourceType: 'module',
23        },
24      },
25      rules: {
26        'no-unused-vars': ['error', { varsIgnorePattern: '^[_A-Z_]' }],
27      },
28    },
29  ])
30
```

File 28: frontend/vercel.json

Frontend - vercel.json

```
1  {
2    "rewrites": [
3      { "source": "/(.*)", "destination": "/index.html" }
4    ]
5  }
6
```

File 29: frontend/package.json

Frontend - package.json

```
1  {
2    "name": "frontend",
3    "private": true,
4    "version": "0.0.0",
5    "type": "module",
6    "scripts": {
7      "dev": "vite",
8      "build": "vite build",
9      "lint": "eslint .",
10     "preview": "vite preview"
11   },
12   "dependencies": {
13     "axios": "^1.13.5",
14     "jszip": "^3.10.1",
15     "react": "^19.2.0",
16     "react-dom": "^19.2.0"
17   },
18   "devDependencies": {
19     "@eslint/js": "^9.39.1",
20     "@types/react": "^19.2.7",
21     "@types/react-dom": "^19.2.3",
22     "@vitejs/plugin-react": "^5.1.1",
23     "eslint": "^9.39.1",
24     "eslint-plugin-react-hooks": "^7.0.1",
25     "eslint-plugin-react-refresh": "^0.4.24",
26     "globals": "^16.5.0",
27     "vite": "^7.3.1"
28   }
29 }
30
```